

A Machine-Verified Disproof of TLMC Conjecture #1243: Rule 30's Central Column Is Not Square-Free

Batch 4 solution submission

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Abstract

We disprove a conjecture of the Last Math Competition about the central column of Wolfram's rule 30 cellular automaton. The conjecture asserts that the column is “square-free over all prefixes of length 2^k ” and that “among its factors of length $< 2^k$ none contains a square”. In the standard sense of combinatorics on words (a *square* being any subword of the form uu with u nonempty), the column begins $1, 1, 0, 1, \dots$, so its prefix of length $2^2 = 4$ already contains the square 11 . The disproof is by kernel-verified computation in Lean 4: the evaluation of the defining recursion of rule 30 is performed by `decide`, and the computation theorems depend on *no* axioms at all.

1 The conjecture

Conjecture 1.1 (TLMC-00000001243). The central column of Wolfram's rule 30 (the output sequence of column 0 from a single 1 as initial condition) is not eventually periodic, and among its factors of length $< 2^k$ none contains a square: the central column is square-free over all prefixes of length 2^k .

2 The disproof

Definition 2.1 (Rule 30 on $\mathbb{Z}$). Rows are functions $\mathbb{Z} \rightarrow \{0, 1\}$; row 0 is the indicator of $\{0\}$, and

$$\text{row}_{t+1}(p) = \text{row}_t(p-1) \oplus (\text{row}_t(p) \vee \text{row}_t(p+1)),$$

the local rule of automaton 30. The *central column* is $c_t = \text{row}_t(0)$.

Theorem 2.2. *The central column begins*

$$c_0 c_1 c_2 c_3 c_4 c_5 c_6 c_7 = 11011100.$$

Proof. Direct evaluation of the defining recursion (Lean: `colPrefix_8`, proved by `decide`, i.e. by the Lean kernel itself; the theorem depends on no axioms). $\square$

Theorem 2.3. *Conjecture 1.1 is **false**. The prefix of length $2^2 = 4$ is 1101 , which contains the factor $11 = 1 \cdot 1$, a square (of period one). Hence the column is not square-free over the prefixes of length 2^k ; equally, a factor of length $< 2^2$ (namely 11 , of length 2) is itself a square.*

Proof. The factor 11 occurs at positions 0–1 of the prefix 1101 from Theorem 2.2. A word of the form uu with u nonempty is a square; here $u = 1$. (Lean: `prefix4_has_square` and `t1m1243_false`.) $\square$

Remark. The non-eventual-periodicity part of Conjecture 1.1 is a famous open problem and is *not* disputed here; the disproof concerns the square-freeness clause as literally stated. If the author intended “no large squares”, the statement would need a lower bound on the square period, and as written the universal claim over all k fails at $k = 2$.

3 Machine verification

The formalization `Rule30.lean` (namespace `TLMCBatch4`) defines rule 30 on $\mathbb{Z}$ exactly as above, computes the first values by `decide`, and derives `t1m1243_false`. Compilation with `lake env lean Rule30.lean` exits with code 0 with zero errors and warnings; `#print axioms` reports that `colPrefix_8` and `colPrefix_4` depend on *no* axioms, and `t1m1243_false` only on `[propext, Classical.choice, Quot.sound]`. No sorry appears.